AR Glasses: The Next Frontier of Computational Photography

Architecture, Applications, and the Future of Spatial Computing



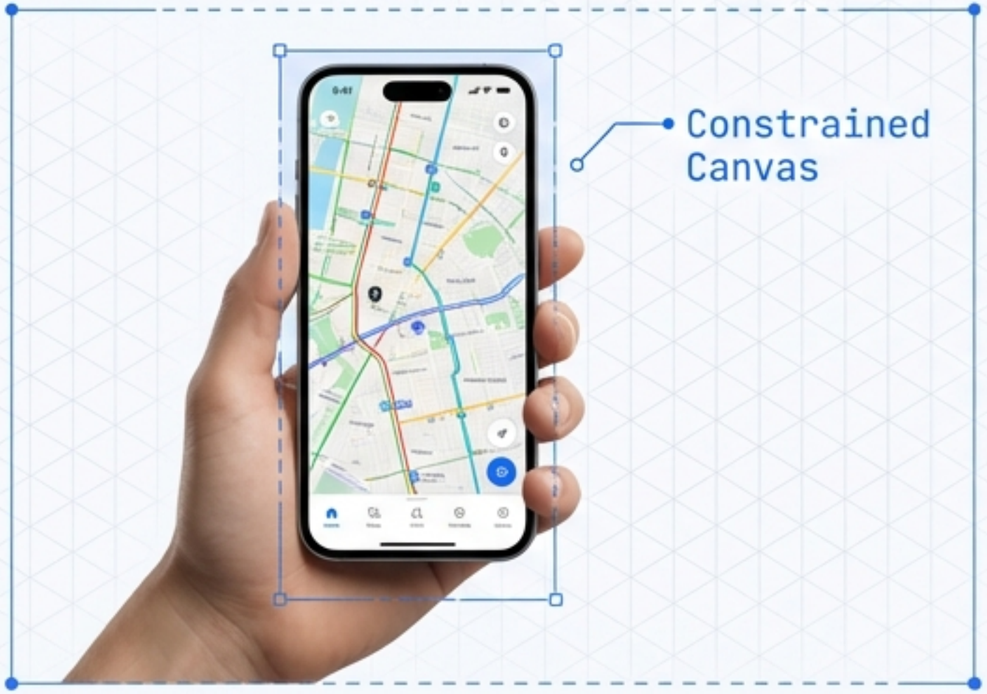
Based on Chapter 25 of 'Mobile Computational Photography'

NotebookLM

Defining the Paradigm Shift

From ‘Looking At’ Screens to ‘Looking Through’ Them

Mobile Era



Spatial Era



Immersive

Digital objects interact with physical geometry.

GEOMETRY_MAPPING = TRUE | LATENCY < 10MS

Hands-Free

Interaction via gaze, gesture, and voice.

MODALITIES: GAZE, GESTURE, VOICE | PROCESSING = EDGE AI

Connected

Constant data stream for real-time assistance.

LIVE_STREAM = 56/66 | BANDWIDTH > 1GBPS | REALTIME_CONTEXT = ON

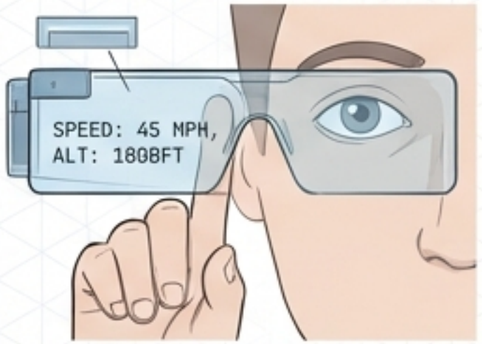
NotebookLM

The Evolution from Concept to Reality

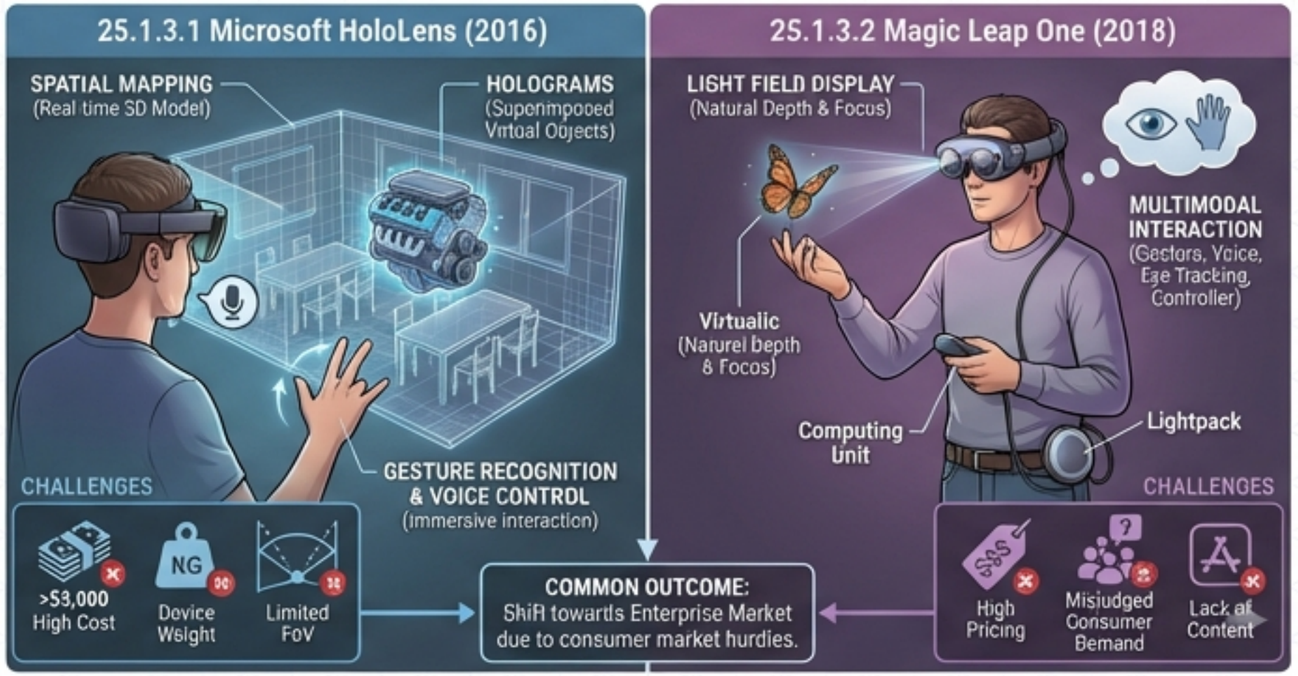
2000s — 2016–2018 — Present

Early Concepts

Industrial HUDs.
Simple data overlays.
No spatial awareness.



Breakthroughs



Modern Era

Miniaturization.
Waveguide optics.
Multimodal interaction.



NotebookLM

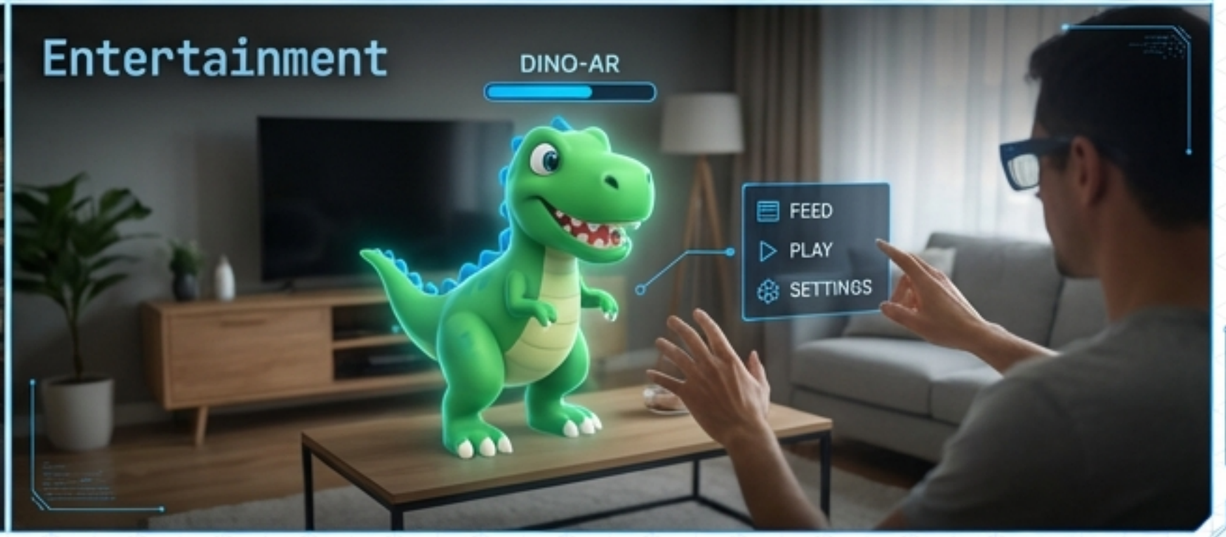
A Multi-Billion Dollar Horizon: Market & Applications

Projected exponential growth (CAGR) through 2030, transitioning from enterprise prototypes to mass-market utilities.









NotebookLM

The Hardware Barrier: The ‘Triangle of Pain’

Balancing physics, power, and form factor.

Battery Life

Real-time rendering drains power; large batteries add weight.



Form Factor

Must be lightweight (<80g) and thermally cool on the face.





Display Quality

High brightness for outdoor use vs. wide Field of View.



Form Factor

Must be lightweight (<80g) and thermally cool on the face.

Display Quality

High brightness for outdoor use vs. wide Field of View.

NotebookLM

Ecosystem, Privacy, and Social Acceptance

Software Ecosystem

- Lack of “Killer Apps”.
- Fragmented OS standards.
- Immature developer tools.



Privacy & Ethics

- Continuous surveillance concerns.
- Biometric security risks (iris/gait).





Market Acceptance

- High entry price (>$3,000).
- Social stigma (“Glasshole” effect).

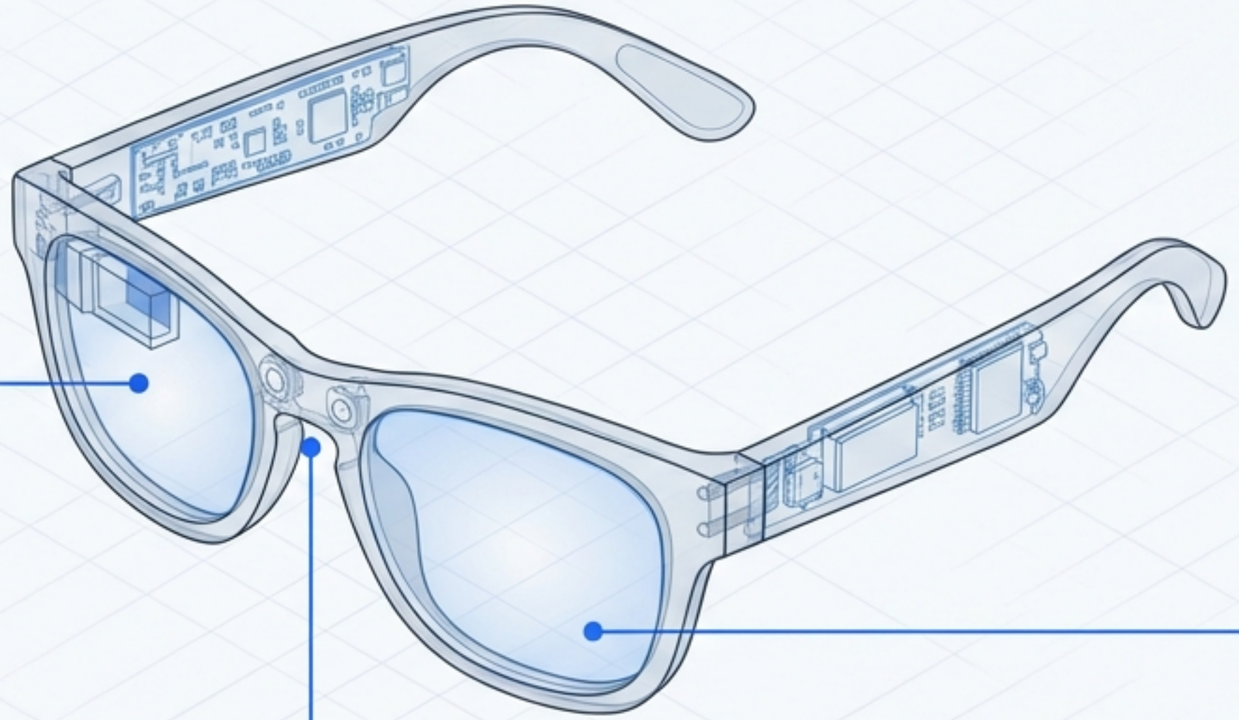




NotebookLM

Hardware Architecture: The Optics of Immersion

The Lens is the Screen



MicroLED

Self-emissive, high brightness, compact. The ‘Holy Grail’ of AR.

Waveguide

Guides light via Total Internal Reflection (TIR). Enables transparent, thin lenses.

LCoS

Liquid Crystal on Silicon. Cost-effective, requires external light source.

NotebookLM

Hardware Architecture: The Sensor Array

The sensor suite enables precise spatial awareness, interaction, and environmental understanding.



Cameras:
RGB + Depth/ToF (World Facing).
IR Eye-Tracking (Inward Facing).

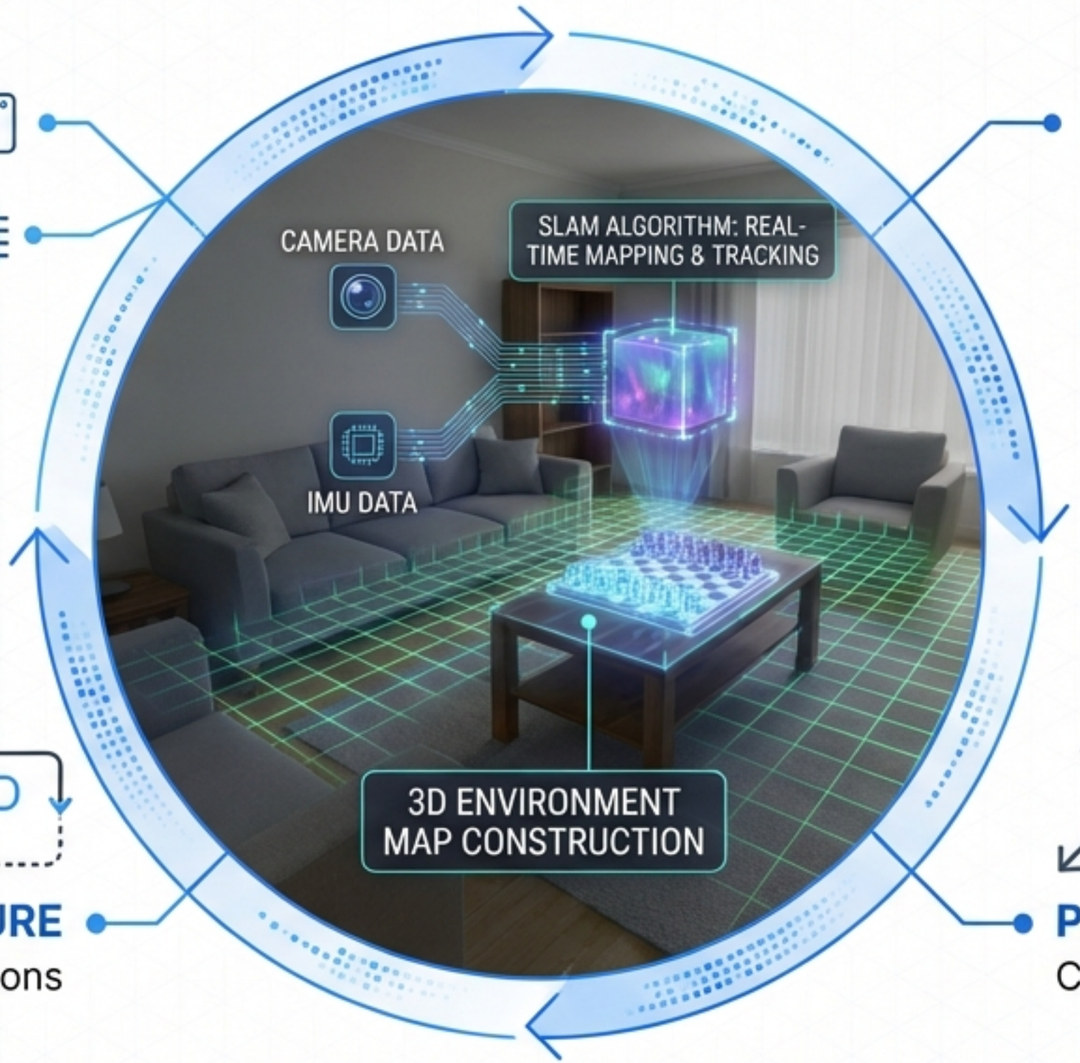
IMU:
Accelerometers & Gyroscopes for low-latency head tracking.

Audio:
Microphone arrays & Spatial Audio drivers.

NotebookLM

The Software Brain: SLAM and Spatial Perception

Simultaneous Localization and Mapping



SENSOR FUSION

Camera Visuals –

IMU Motion –

FEATURE EXTRACTION

Identifying corners/edges

LOOP CLOSURE

Correcting drift/Recognizing locations

POSE ESTIMATION

Calculating 6DoF position

NotebookLM

Connectivity: The World is the Computer



Remote Rendering
Stream high-fidelity 3D assets.

Shared Spaces
Multi-user synchronization.

Data Access
Instant database retrieval.

Offloading processing reduces weight and heat on the device.

NotebookLM

Case Study: Apple Vision Pro

The “Spatial Computing” Approach

Dual-Chip Architecture

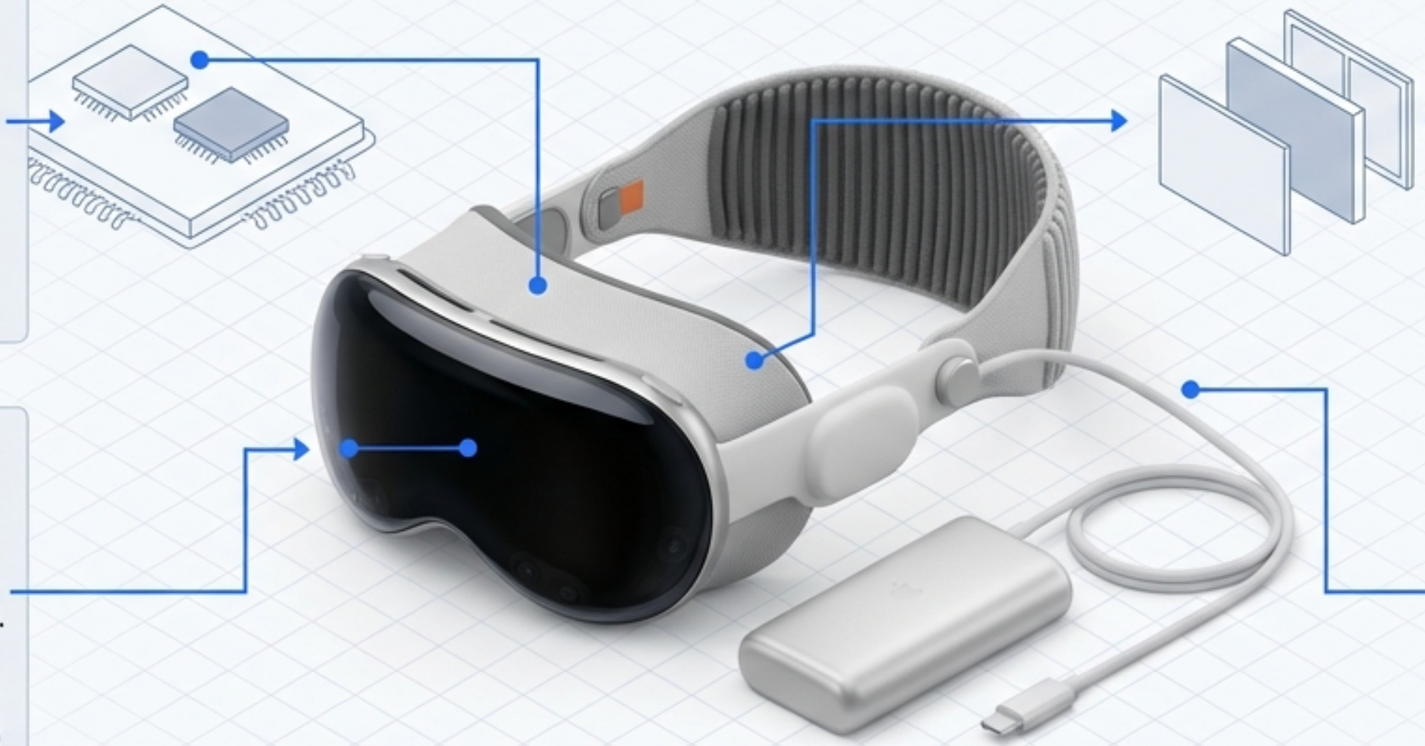
M2 (Compute) + R1 (Sensor Processing).

(M2 Main, R1 Sensor Data) - Low Latency

Display

Micro-OLED, >4K per eye. Video Passthrough.

(>4K per eye) - Brightness, Contrast, Color



Interaction

Eye tracking + Hand gestures. No controllers.

(Foveated Rendering, Interaction; Controller-free, Cameras/Sensors)

Ecosystem

Seamless integration with iCloud/Mac services.

(Service Integration, iCloud, Mac, visionOS

IMMERSIVE AUDIO (Surround Sound) — **visionOS** — **SPATIAL AUDIO** — **INDEPENDENCE** (Standalone Device - Power/Heat Challenge) — **TARGET AUDIENCE** (High-end Consumers, Pros, Developers, Early Adopters) — **EXORBITANT PRICE** (Starting) — **HARDWARE SALES** (Primary Revenue) — **APP STORE** (App Sales, In-app Purchases) — **ECOSYSTEM DEVELOPMENT** (Attracting Developers) — **SERVICE INTEGRATION**

NotebookLM

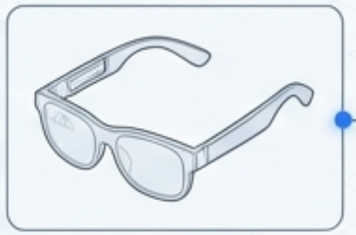
Case Study: Meta (Quest & AR)

Building the Metaverse

Self-Developed Chips
(Optimized Performance & Efficiency)

Advanced Display Technology
(MicroLED, Waveguides)













Hardware

Custom Silicon.
Project Nazare (True AR).
Waveguides & MicroLED.

Technical Data:
JetBrains Mono, Electric Blue

Meta Ecosystem

JetBrains Nono, Electric Blue

Software

Horizon OS.
Social Presence focus.

Technical Data:
JetBrains Nono, Electric Blue



Meta Reality Labs
(CV, SLAM, Rendering, AI)







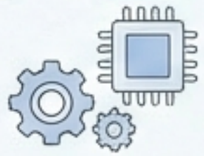
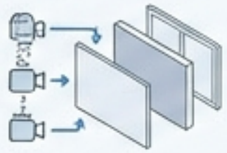
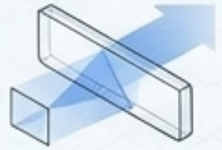
Interaction

Hand tracking + Neural interfaces (EMG).

JetBrains Meno, Electric Blue

NotebookLM

Diverging Paths: Apple vs. Meta

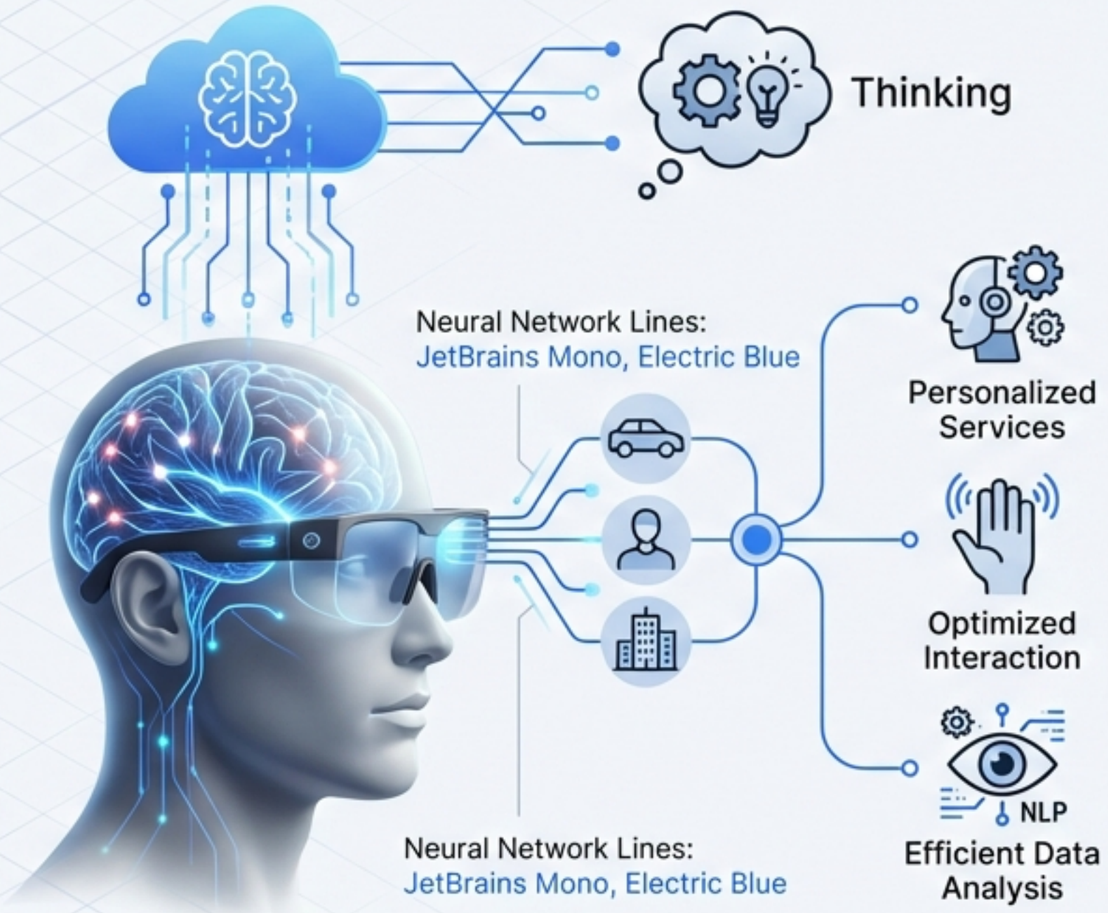
	Apple (Vision Pro)	Meta (Quest / AR)
Focus	Productivity & Computing 	Social & Metaverse 
Target	Pro / High-End 	Mass Market / Gamers 
Display	Video Passthrough (Opaque) 	Optical See-Through (Transparent) 
Ecosystem	Closed / Integrated 	Open / Social 

Apple bets on hardware perfection today.
Meta bets on social ubiquity tomorrow.

NotebookLM

The Horizon: When AI Meets XR

From Displaying Data to Understanding Context





- **Intelligent Perception**

  Proactive assistance (e.g., identifying plants, translating menus).



- **Natural Interaction**

  LLM-driven voice interfaces replace menus.



- **Generative AR**

  On-the-fly 3D content creation.

NotebookLM

Conclusion: The Era of Invisible Computing

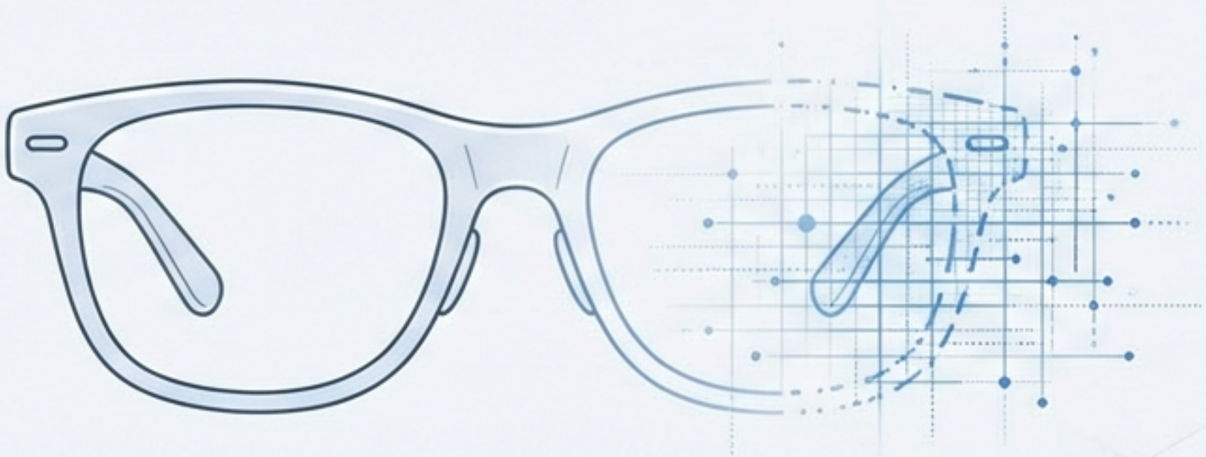
AR glasses represent the culmination of mobile computational photography—fusing advanced optical hardware, SLAM algorithms, and AI to merge the digital and physical worlds.

Short Term

Passthrough headsets and smart glasses.

Long Term

True, transparent, all-day AR replaces the smartphone.



NotebookLM